Migration Surge Prediction: An Early Warning System

Ibrahim	**Haseeb**	**Asadohoo**
58776310	58791350	58452930
Sameer	**Ayaan**	**Pak Hei Ng**
58771398	58970130	58650183

SDSC 2005 – Introduction to Computational Social Science
Group 10 Final Report
April 2026

Abstract

Governments and humanitarian organisations are routinely caught off-guard by sudden spikes in migration, yet the underlying signals, including shifting news coverage, search behaviour, and macroeconomic volatility, are observable months in advance. This study presents a multi-modal early warning system that synthesises over 170,000 news articles, Google Trends search-intent data, and Real Effective Exchange Rate (REER) indicators across 15 high-volume origin countries over a nine-year observation window (2017–2026). We use these modalities to (i) identify leading signals of migration pressure and (ii) forecast surges in legal immigration (visa issuances) at one-to-six-month horizons; border encounters are incorporated as a complementary reference series to contextualise dynamics across formal and informal migration channels. While search-intent signals are informative in correlation and time-series analyses, they are not included in the final predictive feature matrix of the reported surge-forecasting models. We evaluate four architectures: a GPU-accelerated Random Forest baseline, a recurrent LSTM with a novel dual-objective *SurgeJointLoss*, a Transformer encoder, and a Horizon-Aware Ensemble that dynamically re-weights models by forecast lead time. Evaluated strictly out-of-time on walk-forward splits (training $\leq$ 2022, testing 2023+), the ensemble achieves an average F1 score of 0.915 with recall of 0.943. Across horizons, F1 ranges from 0.96 at Lead 1 to 0.86 at Lead 6, while recall remains at or above 0.92, so fewer than one in ten genuine crises goes undetected. These results demonstrate that fusing heterogeneous digital trace data with purpose-built deep learning can transform migration forecasting from a reactive exercise into a proactive planning tool.

Contents

1 Introduction **3**
1.1 Study Context and Motivation . . . 3
1.2 Research Questions . . . 3

2 Methods **3**
2.1 Study Design and Data Scope . . . 3
2.2 Data Acquisition and Source Hierarchy . . . 3
2.3 NLP Processing and Event Discovery . . . 4
2.4 Exploratory Data Analysis . . . 6
2.5 Lead–Lag Signal Analysis . . . 8
2.6 Predictive Models . . . 8
2.7 Validation Strategy . . . 9

3 Results **9**
3.1 Multi-Signal Analysis (RQ3) . . . 9
3.2 Model Performance . . . 10
3.3 Architecture Insights (RQ2) . . . 11

4 Conclusion **11**
4.1 Practical Implications . . . 11
4.2 Limitations . . . 12
4.3 Future Work . . . 12
4.4 Code and Data Availability . . . 12

5 Contribution Statement **13**

Appendices **16**

A Data and Pipeline Engineering (Reproducible Overview) **16**

B Panel Schema and Feature Dictionary **17**

C Per-Horizon Surge Classification **17**

D Ensemble Weight Schedule **18**

E Detailed Event Signal Analysis **18**

F Cluster Quality Analysis **19**

G F1-Score Trajectory Across Forecast Horizons **20**

H Future Work **21**

1 Introduction

1.1 Study Context and Motivation

Global migration is driven by a deeply intertwined web of push and pull factors, including geopolitical instability, economic opportunity, conflict, climate events, and family reunification, making the timing and magnitude of migration flows inherently difficult to anticipate. For the roughly 280 million international migrants worldwide [1], the decision to move is rarely sudden; it is incubated by months of deteriorating conditions at home and perceived opportunity abroad. The signals of that incubation period, such as a spike in "US asylum" searches, a currency devaluation, or a surge in conflict-related news coverage, are increasingly observable in real time through digital trace data.

Yet current migration management remains overwhelmingly reactive. Governments confront surges only *after* they materialise, scrambling to allocate housing, legal processing capacity, and humanitarian aid when border crossings have already overwhelmed existing infrastructure. In May 2023, U.S. Southwest border encounters peaked at roughly 880,000 in a single month [2], approximately 18 times the 2018–2019 baseline. Traditional forecasting suffers from what we term a **"precision blindspot":** models trained on slow-moving historical indicators adequately predict the normal baseline but systematically miss the extreme spikes that actually trigger crises.

This project proposes an **Early Warning System** that addresses this blindspot by synthesising three families of real-time digital signals: (1) news event coverage reflecting geopolitical conditions in origin countries, (2) Google Trends search-intent data capturing migration-related interest, and (3) macroeconomic exchange-rate volatility, into actionable, multi-horizon surge forecasts spanning one to six months ahead.

1.2 Research Questions

- **RQ1:** How can multi-modal digital trace data (news coverage, search intent, and macroeconomic indicators) be synthesized to predict sudden surges in migration?
- **RQ2:** How does the predictive accuracy of various architectures (Random Forest, LSTM, Transformer) change across forecasting horizons (1–6 months)?
- **RQ3:** Which data modality (news coverage, Google Trends, or exchange rates) serves as the strongest early indicator of a migration spike?

2 Methods

2.1 Study Design and Data Scope

The analytical framework is structured as a monthly, country-level forecasting system that analyzes two operational endpoints: legal visa issuances and unauthorized border encounters. The forecasting task and model evaluation in this report focus on **visa-issuance surges** (a high-quality, consistently measured legal-flow target); encounter data is used to **contextualise** migration dynamics and validate that major disruptions (e.g., post-pandemic changes and 2023 spikes) appear across channels, but it is not the primary supervised target in the surge models reported here. All source streams are harmonized to a calendar-month cadence prior to statistical testing and model training. The final modelling scope is restricted to the 15 highest-volume origin countries (Afghanistan, China, Colombia, Cuba, Dominican Republic, El Salvador, Guatemala, Haiti, Honduras, India, Mexico, Pakistan, Philippines, Venezuela, and Vietnam). This restriction is empirically motivated. Initial exploratory data analysis revealed a strongly Zipfian cross-country concentration, where the top 20% of sending countries account for 88.3% of all visas issued and 81.7% of all surge months. This regularity anchors the scope to avoid signal dilution from a long tail of low-frequency origins (see Section 2.4).

2.2 Data Acquisition and Source Hierarchy

The data ingestion pipeline relies on programmatic retrieval from live sources. Hugging Face repositories are utilized strictly as an archival and reproducibility layer to bootstrap precomputed

artifacts, rather than serving as the methodological origin of the training data.

Ground Truth: Visa Issuances. Monthly immigrant visa issuance statistics were collected from the U.S. Department of State [3], spanning January 2017 through September 2025. The 108 monthly PDF reports were parsed programmatically using PyMuPDF table extraction. The routine normalizes headers, excludes aggregate rows, and maps 118 distinct visa classification codes into seven interpretable categories (family-sponsored, employment-based, special immigrant, humanitarian, returning resident, diversity lottery, and non-immigrant), yielding ~100,000 records.

Ground Truth: Border Encounters. Southwest U.S.–Mexico land border encounter data was obtained from U.S. Customs and Border Protection [2], comprising eight CSV datasets (FY 2019–2026 YTD) merged via Polars [4] lazy evaluation. These metrics represent the unauthorized-flow counterpart to legal visa issuances.

News Articles. A large-scale automated pipeline collected articles via the Google News RSS API [5] across all 15 countries and 8 migration-relevant topics (e.g., "US visa," "US asylum," "CBP One"), producing 120 unique query combinations. Articles were fetched asynchronously with bounded concurrency (16 connections), decoded in batches of 20, and extracted via Trafilatura [6] parallelized across CPU cores. Failed fetches and non-200 status codes were discarded. This yielded **170,784** raw articles, of which **104,333** passed quality filters, achieving a 61.1% success rate. Details of the pipeline architecture appear in Appendix A.

Google Trends. Monthly Google Trends time series were collected for eight migration-intent keywords (`us_visa`, `us_asylum`, `green_card`, `cbp_one`, `usaid`, `united_states_passport`, `us_jobs`, `study_us`) across all 15 countries [7].

Macro-Financial Context. Real Effective Exchange Rate (REER) data was obtained from the IMF [8] for the 6 of 15 countries where data was available (China, Colombia, Dominican Republic, Mexico, Pakistan, Philippines), representing a known coverage limitation.

Table 1 summarises the four data streams, their scope, and the role of each in the modelling pipeline.

Table 1: Summary of data sources used in the study.

Source	Type	Period	Records	Pipeline Role
U.S. Dept. of State	Visa issuances (PDF)	2017–2025	~100,000	Primary target (legal flow)
U.S. CBP	Border encounters (CSV)	2019–2026	Multi-million	Secondary target (informal flow)
Google News RSS	News articles (HTML)	2017–2026	104,333	Event feature; sentiment
IMF REER	Exchange rates (API)	2017–2026	6 countries	Macro feature (partial)
Google Trends	Search interest (API)	2017–2026	120 series	Collected; excluded from features

2.3 NLP Processing and Event Discovery

Embedding Generation. Each of the 104,333 articles was tokenized (max 8,192 tokens) and encoded into a 768-dimensional embedding using a Jina v5 model [9] accelerated via TensorRT INT4 quantisation [10]. TensorRT compilation applied four optimisations: (i) *operator*

fusion, combining adjacent GEMM, bias, and activation operations into a single CUDA kernel; (ii) *layer fusion*, merging consecutive matrix-multiply and non-linear layers; (iii) *CUDA graph capture*, recording the full inference computation graph once to eliminate per-step kernel launch overhead; and (iv) *INT4 weight quantisation* for the embedding model and INT8 (W8A16) for the Flan-T5-Large labelling model. These optimizations yielded throughputs exceeding **20,000 tokens/second** for Jina v5 (sequence length 8,192) and **6,000 tokens/second** for Flan-T5-Large (sequence length 512), enabling overnight batch processing. Quantization validation confirmed mean cosine similarity of 0.9999 against the FP32 baseline. Non-finite vectors were removed prior to clustering.

Event Clustering. To discover thematic event groups, HDBSCAN [11] was applied to the embeddings via NVIDIA cuML [12] (`min_cluster_size=10`, `min_samples=5`, Euclidean metric). To preserve full semantic fidelity, clustering was performed directly on the raw 768-dimensional space without prior dimensionality reduction. The resulting clusters were subsequently projected into lower dimensions via UMAP [13] and t-SNE [14] exclusively for visual diagnostic inspection.

Autonomous Cluster Labelling. A TensorRT-optimized Flan-T5-Large model (INT8 W8A16 quantisation) [15, 10] generated 2–3 word descriptive labels for each cluster from five sampled article headlines using greedy decoding and constrained output length. Approximately 75% of clusters received meaningful labels (e.g., *"Honduras Evangelical Pastors"*, *"Cuba: Trump Policy"*); the remainder fell back to a generic *"Migration event update"* tag when the output failed coherence checks, and these were retained as-is.

Headline-Level Scoring. A lexicon-based scoring function was applied to each article **headline** using a domain-specific vocabulary of 19 positive terms (e.g., "growth," "peace," "jobs") and 27 negative terms (e.g., "crisis," "conflict," "violence"):

$$s = \text{clip}\left(\frac{n_+ - n_-}{\sqrt{n}},\ -1,\ 1\right), \tag{1}$$

where n_+ is the count of matched positive lexicon terms in the headline, n_- is the count of matched negative terms, n is the total token count of the headline, and $\text{clip}(\cdot,\ -1,\ 1)$ clamps the raw score to avoid division-by-small-n inflation. Monthly event sentiment summaries, including mean score and polarity composition, were aggregated at the country-month level; score shocks were contemporaneously flagged via z-score thresholds ($z \leq -1.0$ negative, $z \geq +1.0$ positive).

Panel Construction and Imputation. All modalities were merged into a single country-month panel. Prior to lag construction, missing values were handled modality-specifically: absent visa issuances and news event counts were zero-filled, while missing exchange rate observations were imputed via forward-filling followed by backward-filling within each country to maintain temporal continuity. Autoregressive **lag features** ($t-1$ through $t-6$) were constructed for exactly three modalities (visa volume, exchange rate, and news event count), yielding 18 input features alongside **lead targets** ($t+1$ through $t+6$) of visa volume. Although Google Trends data was collected and analyzed (Section 2.5 and Section 3.1), it was excluded from the final predictive feature matrix used by the reported surge-forecasting models. In our experiments, Trends signals frequently showed statistically significant correlations but translated weakly into out-of-time gains under linear time-series baselines, motivating its use as an *important complementary modality* for interpretation and future model extensions rather than a core feature block in the final feature set reported here. Any rows retaining missing values after lag and lead generation were dropped.

2.4 Exploratory Data Analysis

Dual-Channel Ground Truth. Figure 1 plots both target streams, monthly visa issuances and unauthorized border encounters, on a shared time axis. The two channels exhibit positive long-run co-movement (Pearson $r = 0.438$), but diverge sharply during the COVID-19 disruption (2020–2021) and the 2023 encounter surge, reflecting the distinct institutional channels they represent.

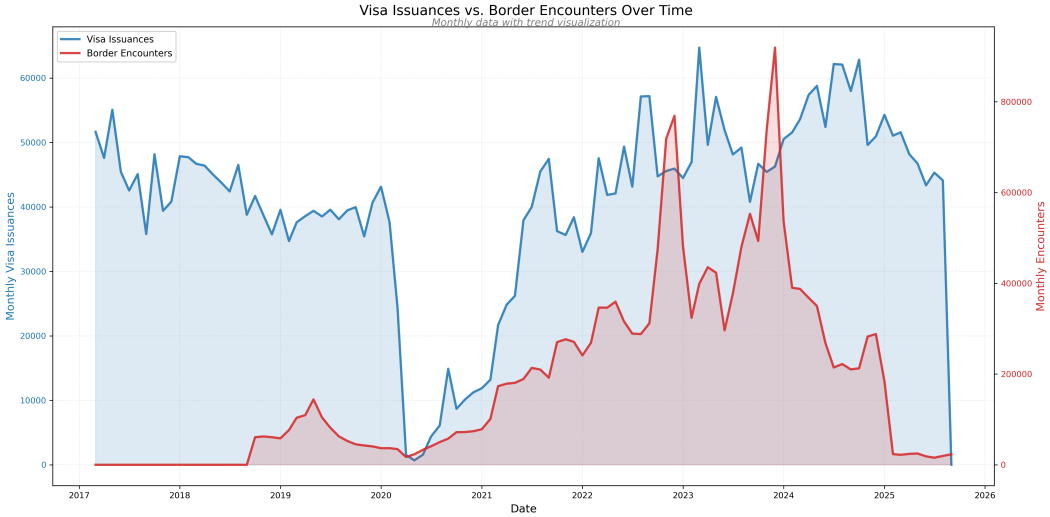

Figure 1: Monthly visa issuances (left axis) and U.S. Southwest border encounters (right axis), 2017–2026. Both series confirm a post-pandemic recovery, but the May 2023 encounter spike (~880,000) dwarfs the visa baseline, illustrating the difference in magnitude between the two migration channels.

Visa Category Composition. Figure 2 shows the decomposition of monthly issuances across the seven visa categories. Family-sponsored visas dominate throughout; employment-based visas grew substantially post-2021. Humanitarian and diversity-lottery categories contribute smaller but volatility-prone shares, making them disproportionate contributors to surge months.

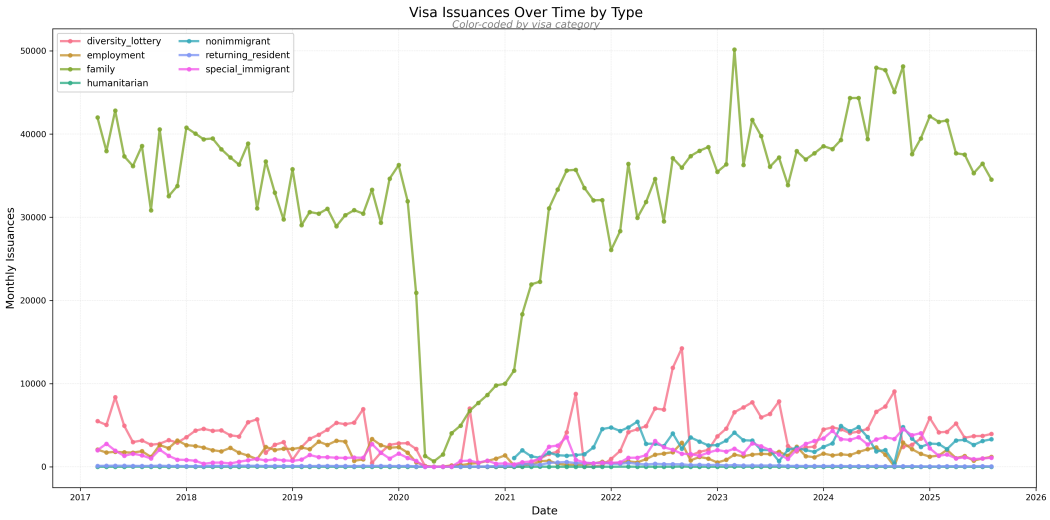

Figure 2: Monthly U.S. visa issuances by category (2017–2025). Family-sponsored visas account for the majority of volume; employment-based and humanitarian flows display distinct post-pandemic trajectories.

Seasonal Patterns and Pandemic Disruption. Figure 3 shows the monthly visa issuance heatmap from 2017 to 2025. Volume was broadly consistent across 2017 to 2019, collapsed during 2020 owing to pandemic-era consulate closures, and rebounded sharply from 2021 onward.

The summer and early-autumn months (July to October) of 2023 to 2024 represent the peak issuance period, confirming a seasonal cycle requiring architectural accommodation.

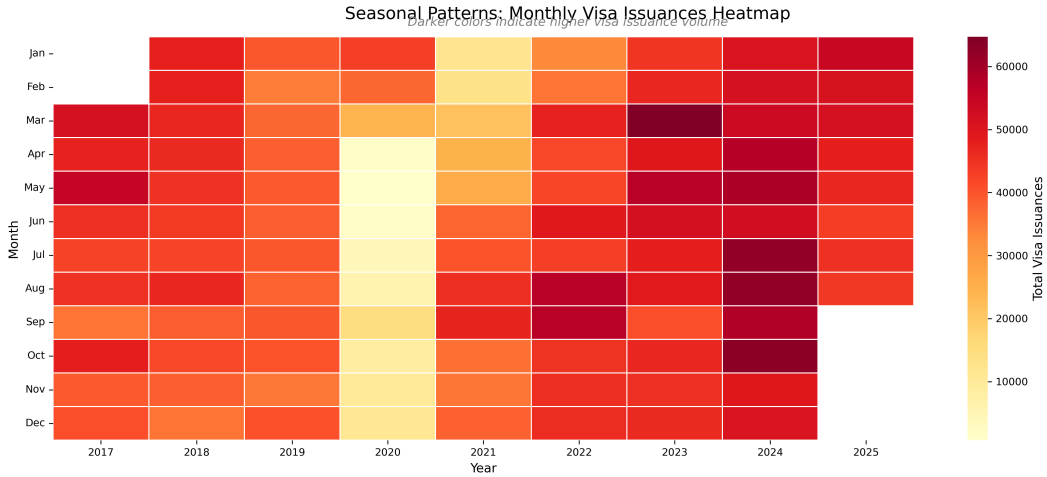

Figure 3: Seasonal Patterns: Monthly Visa Issuances Heatmap (2017 to 2025). Darker red shading indicates higher monthly volume. The light-coloured column at 2020 reflects near-complete consulate closures during the COVID-19 pandemic.

Zipfian Concentration and Country Volatility. As noted in Section 2.1 and illustrated in Figure 4, the cross-country distribution of visa issuances follows a Zipfian law: the top 20% of sending countries contribute 88.3% of all issuances, and 81.7% of all surge months.

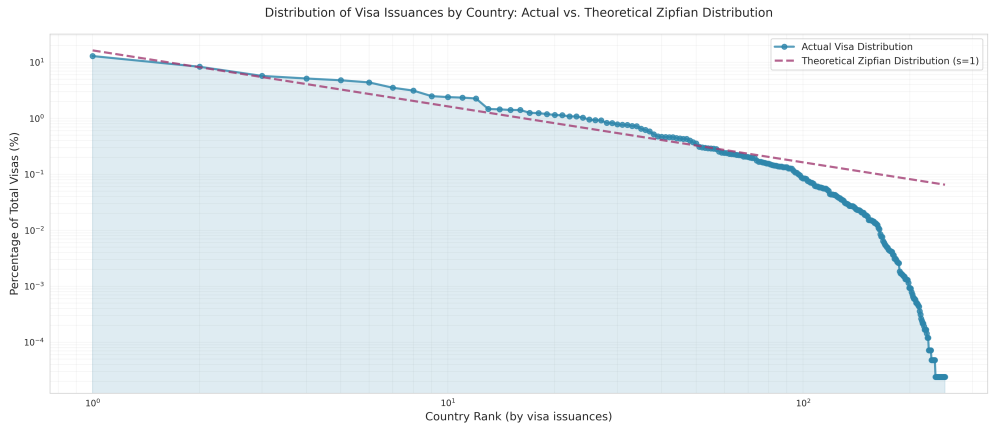

Figure 4: Zipfian distribution of U.S. visa issuances by origin country. The highly skewed concentration justifies the methodological scope limitation to the top 15 highest-volume countries, which account for the vast majority of baseline volume and surge events.

Country-Level Volatility. Country-level volatility, measured by the coefficient of variation (CV) of monthly issuances, identifies surge-prone origins. As Figure 5 shows, Cuba (CV 2.89), Afghanistan (CV 2.81), and Pakistan (CV 2.48) are the most volatile origins, while the Dominican Republic (CV 1.55) is comparatively stable despite providing strong lead signals.

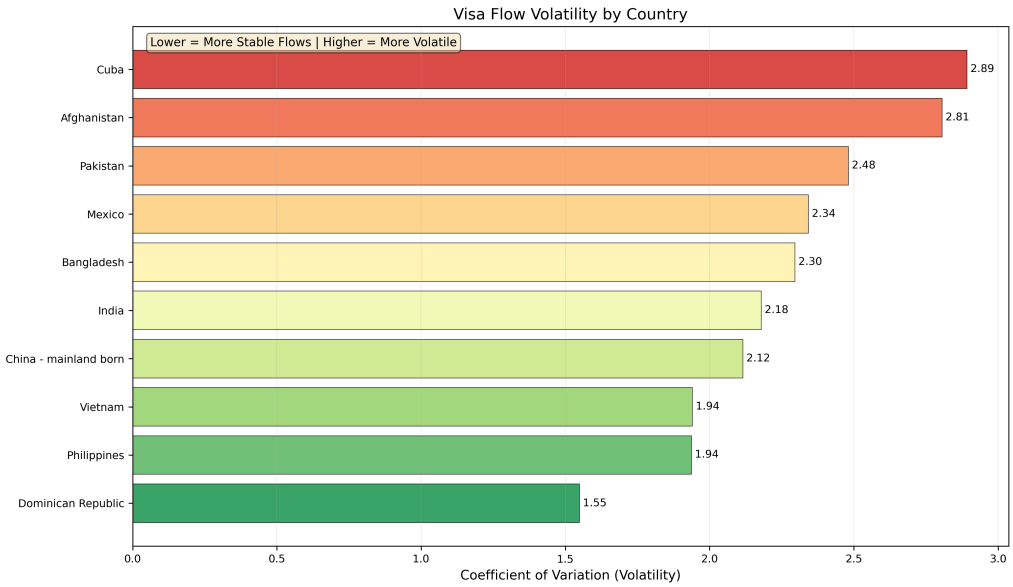

Figure 5: Visa flow volatility by country, measured as the coefficient of variation of monthly visa issuances. Higher values indicate more erratic, surge-prone flows. Colour encodes volatility level (red = high, green = moderate).

2.5 Lead–Lag Signal Analysis

Pearson correlation was computed for each country-signal-lag combination (lags 0 to 6) against visa volume, with statistical significance controlled via Benjamini-Hochberg FDR [16] at an adjusted p-value threshold of $q \leq 0.10$. For **signal screening** in this section, "surge months" were defined by a dual heuristic criterion: volume above the 75th percentile *or* month-over-month growth exceeding 30%. For **model evaluation** (Section 2.7), we use a single operational surge label based on a rolling mean/standard-deviation threshold ($> \mu + 1.5\sigma$) to standardize classification across countries and horizons. Signal quality was explicitly quantified via surge precision and surge lift. Event analysis yielded 392 country-label-lag combinations, with 58 surviving the false-discovery filter. The strongest positive event signal was observed for Honduras, where an evangelical-pastors label at a 2-month lag achieved a correlation of 0.4250 ($q = 0.0114$) and a surge lift of 1.2291. Exchange-rate coverage was more limited, yet the Dominican Republic exhibited the strongest macro relationship, with a correlation of 0.4980 at a 2-month lag. These findings dictate the integration of events and exchange rates as auxiliary features rather than standalone predictors. A key directional pattern, consistent across countries and modalities, is that **negative sentiment and macro shocks lead visa increases by 2–4 months**: deteriorating conditions in origin countries function as push factors that incubate emigration intent before it materialises as formal applications.

2.6 Predictive Models

Input modalities were conditioned according to the architectural inductive biases. For the sequential neural architectures (LSTM and Transformer), the 18 input features were Z-score normalized utilizing a standard scaler fitted exclusively on the training distribution. Conversely, the tree-based baseline operated directly on the unscaled tabular features.

cuML Random Forest. A GPU-accelerated Random Forest [17, 12] with 50 estimators and maximum depth 6 served as a non-temporal tabular baseline. Six independent models were trained, one per horizon, to capture lead-specific structures independently.

LSTM with SurgeJointLoss. A two-layer LSTM [18] (hidden dimension 64) with learned country embeddings (dimension 8) processed the six-step sequence input. The model was trained

using the Adam optimizer [19] with a learning rate of 10^{-3} and a batch size of 256 for 25 epochs. Training optimized a novel dual-objective loss function, *SurgeJointLoss*:

$$\mathcal{L} = \alpha\ \mathcal{L}_{\text{Huber}}(\hat{y},\ y)\ +\ (1-\alpha)\ \mathcal{L}_{\text{BCE}}\Big(\hat{y}-\tau,\ \mathbf{1}\big[y\ >\ \mu + 1.5\,\sigma\big]\Big), \tag{2}$$

where $\hat{y}$ is the predicted monthly visa volume, y is the observed volume, $\alpha = 0.6$ is a fixed mixing weight placing 60% emphasis on volume regression and 40% on surge detection, τ is a learned discrimination threshold, μ is the rolling mean of historical visa volume, σ is its rolling standard deviation, $\mathcal{L}_{\text{Huber}}$ is the Huber regression loss [20], and $\mathcal{L}_{\text{BCE}}$ is Binary Cross-Entropy. The deterministic surge boundary is defined at $\mu + 1.5\sigma$. Both neural architectures were implemented in PyTorch [21].

Transformer Encoder. A two-layer Transformer [22, 21] utilized a linear input projection ($d_{\text{model}} = 64$), 4 attention heads, and learned positional encodings over the six-month window. It was trained with Mean Squared Error loss for 20 epochs using the Adam optimizer (learning rate 10^{-3}, batch size 256) as a pure attention-based comparator.

Horizon-Aware Ensemble. The final model blended predictions from all architectures utilizing hard-coded, lead-specific weights that dynamically shift mass from the Random Forest and LSTM at short horizons toward the Transformer at extended horizons, optimally combining inductive biases (see Appendix D for the full schedule and precise weights).

2.7 Validation Strategy

All models were evaluated under a strict **walk-forward, out-of-time (OOT)** protocol: training was restricted exclusively to observations through December 2022, and evaluation was performed on 2023 and beyond. To strictly prevent target leakage, model training utilized fixed-epoch loops without a defined validation hold-out split or early stopping mechanisms. A month-country observation was classified as a surge if visa volume exceeded 1.5σ above the rolling mean, i.e., $y > \mu + 1.5\sigma$ (yielding approximately 103 to 114 surge instances in the test set, depending on horizon-induced row dropping). Performance was quantified via RMSE for volume prediction, alongside Precision, Recall, and F1 for surge detection. The benchmark revealed that the LSTM excels in short-horizon RMSE, the Random Forest remains competitive for near-term classification, and the Transformer enhances recall at longer horizons. The horizon-aware ensemble balances these complementary strengths across the entire 1 to 6 month forecast window.

3 Results

3.1 Multi-Signal Analysis (RQ3)

News Event Signals. Of 392 country–event–lag combinations tested, 58 (14.8%) achieved significance ($q \leq 0.10$). The strongest single signal was the Dominican Republic, where the "First Presidency" sentiment cluster showed $r = 0.617$ at a 3-month lag ($q = 3.3 \times 10^{-9}$), reflecting how political events in the DR precede visa application surges by 3 months. Cuba's "Trump Policy" cluster showed $r = -0.595$ at lag 0 ($q = 3.4 \times 10^{-9}$), reflecting immediate U.S. policy impacts. Additional cluster signals include Guatemala's "Border Policy" cluster ($r = 0.523$, lag 3) and Venezuela's "Economic Crisis" cluster ($r = 0.509$, lag 5), consistent with political and macroeconomic push factors driving departure timelines. Eleven of 15 countries exhibited at least one significant event signal. Full results appear in Appendix E.

Exchange Rate Signals. Among 6 countries with REER data, 23 of 42 lag rows (54.8%) were significant. The Dominican Republic dominated ($r = 0.498$, lag 2, $q = 2.6 \times 10^{-6}$). Overall surge lift was modest (mean 1.009), confirming exchange rates as a complementary rather than standalone signal.

Google Trends. Correlations were significant for 66 of 120 rows (55.0%), but a trends-enhanced VAR [23, 24] model *underperformed* the autoregressive baseline (−5.97% mean RMSE for visas), indicating that translating keyword correlations into out-of-sample gains requires non-linear modelling.

Modality Ranking. News event coverage emerged as the strongest individual predictor (highest single-signal correlations, most country-specific leads); exchange rates added meaningful signal where available (6 countries); Google Trends showed the broadest coverage but the weakest out-of-sample translation, which motivates the multi-modal fusion approach.

3.2 Model Performance

Table 2 presents the aggregate surge-classification performance across all six horizons.

Table 2: Average surge-classification performance (Leads 1–6, OOT test set).

Algorithm	Precision	Recall	F1 Score
cuML Random Forest	0.870	0.943	0.907
LSTM (SurgeJointLoss)	0.895	0.820	0.855
Transformer Encoder	0.892	0.917	0.905
Horizon-Aware Ensemble	**0.887**	**0.943**	**0.915**

The ensemble achieves the highest average F1 (0.915) by combining the RF's high recall (0.943) with the Transformer's balanced precision–recall profile. The LSTM's conservative SurgeJointLoss objective yields the highest precision (0.895) at the cost of lower recall (0.820). Per-horizon breakdowns are provided in Appendix C.

Table 3 reports volume prediction accuracy (RMSE) for the three base models.

Table 3: Out-of-time RMSE by model and horizon (lower is better). Bold marks best per row.

Horizon	Random Forest	Transformer	LSTM
Lead 1	298.79	303.43	**277.40**
Lead 2	341.79	343.61	**320.77**
Lead 3	351.55	359.20	**346.03**
Lead 4	**364.01**	381.20	381.00
Lead 5	428.35	407.67	**404.23**
Lead 6	**401.38**	426.48	424.59

The LSTM dominates near-term volume prediction (Leads 1–3) by 10–20 RMSE points; at longer horizons the RF maintains competitive parity.

Precision–Recall Trade-offs. Figure 6 plots the precision–recall profiles of all four models (averaged over Leads 1–6), offering a complementary view of surge-detection trade-offs.

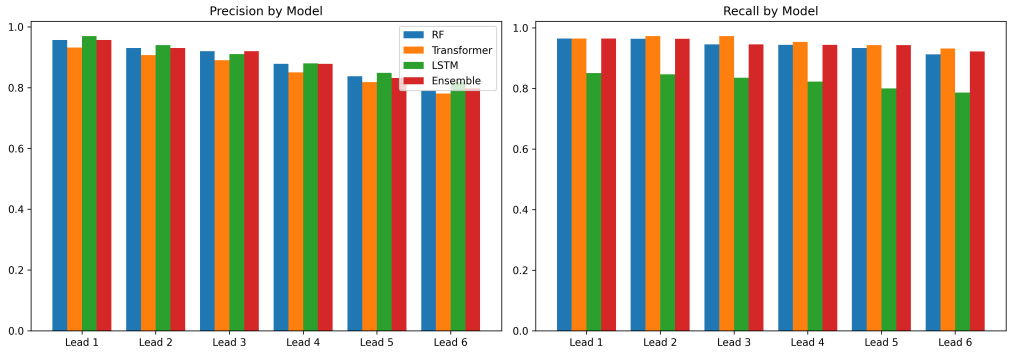

Figure 6: Precision–recall comparison across all four architectures (OOT, Leads 1–6 average). The ensemble (star) and Random Forest achieve the best recall (>0.94); the LSTM (triangle) trades recall for precision via the SurgeJointLoss objective.

3.3 Architecture Insights (RQ2)

1. **Random Forest:** highest single-model F1 at Lead 1 (0.97) and recall of 0.943 on average, but precision degrades at longer horizons.
2. **LSTM:** highest precision (0.895 average) thanks to SurgeJointLoss; when it fires an alarm, it is the most likely to be genuine. The trade-off is lower recall (0.820).
3. **Transformer:** superior long-horizon recall (90% at Lead 6) and the highest individual-model F1 at Lead 6 (0.87), capturing slow geopolitical dynamics via self-attention.
4. **Ensemble:** by shifting trust from RF/LSTM at short leads toward the Transformer at long leads, it achieves the best F1 at every horizon and maintains **recall** $\geq$ 0.92 across all six months.

4 Conclusion

This study has demonstrated that multi-modal digital trace data, when processed through architectures designed to capture surge dynamics, can reliably anticipate migration crises months before they materialise. Across all three research questions the evidence converges on a clear answer: fusing news event coverage and macroeconomic indicators into a Horizon-Aware Ensemble yields consistent gains over every single-modality or single-architecture baseline, with the ensemble achieving an average F1 of 0.915 and recall $\geq$ 0.92 at every forecast horizon from one to six months. These results indicate that the "precision blindspot" afflicting traditional reactive migration management is not an inherent feature of the forecasting problem, but an artefact of relying on slow-moving aggregate statistics rather than the rich signal landscape now observable through digital sources. Pre-emptive planning, including staffing adjustments, logistics pre-positioning, and diplomatic engagement, becomes tractable when high-recall early warnings arrive months in advance.

4.1 Practical Implications

- **Lead 1 (F1 0.96):** Immigration processing centres can adjust staffing for the coming month with high confidence.
- **Lead 3 (F1 0.93):** Humanitarian organisations can pre-position supplies and shelter capacity in anticipated surge corridors.
- **Lead 6 (F1 0.86):** National-level budgetary and diplomatic planning can be initiated with sufficient institutional lead time.

The finding that news event signals around U.S. policy announcements serve as the strongest leading indicators suggests that **monitoring origin-country media coverage could function as a low-cost early warning input** even without sophisticated modelling infrastructure.

4.2 Limitations

The Digital Divide. Google Trends and digital news only reflect populations with internet access. The most vulnerable migrants including those displaced by conflict or extreme poverty in regions with limited digital infrastructure may be systematically underrepresented, creating a blind spot for the most at-risk groups.

Correlation, Not Causation. This is a *predictive* model, not a *causal* one. Lead–lag correlations identify temporal associations, but establishing causal mechanisms would require model interpretability outside the scope of the project.

Exchange Rate Data Gap. REER data was available for only 6 of 15 target countries [8], excluding high-priority origins (Afghanistan, Cuba, Haiti, Honduras, Venezuela, among others).

Black Swan Events. Genuinely novel shocks lie beyond the model's anticipatory reach. Events such as sudden law changes, natural disasters and unprecedented policy events are not anticipated by the model.

Google Trends as a Weak Standalone Predictor. Despite significant keyword-level correlations, the trends-enhanced VAR model underperformed the autoregressive baseline (−5.97% RMSE on average), indicating that search-intent signals require non-linear architectures to be leveraged effectively.

News Coverage Bias. The 61.1% pipeline success rate means roughly 39% of articles were lost to extraction failures, paywalls, or anti-scraping measures. Coverage is skewed toward English-language media, potentially underrepresenting local-language reporting.

Model Staleness and Distributional Drift. The model is anchored to the 2017–2022 training window. Without periodic retraining, the system could become obsolete.

Survivorship Bias in Encounter Data. Border encounters record only detected crossings. Migrants who evade apprehension are absent. The model therefore learns detection rates rather than the true migration pressure.

4.3 Future Work

Key extensions include social media integration for finer-grained intent signals, cross-country transfer learning from data-rich to data-sparse origins, real-time streaming deployment, and causal discovery via Granger causality or structural equation models. See Appendix H for the full enumeration.

4.4 Code and Data Availability

The full codebase, including data-collection scripts, model implementations, and reproduction notebooks, is publicly available at `https://github.com/ibitec7/migration`. Pretrained model weights, processed datasets, and intermediate artifacts (embeddings, cluster labels, production outputs) are hosted on Hugging Face at `https://huggingface.co/sdsc2005-migration`.

5 Contribution Statement

This project was completed collaboratively. Table 4 summarises contributions at a high level. Checkmarks indicate meaningful contribution within each component area.

Table 4: Contribution matrix by project component.

Name	Data	Proc.	NLP	Model	Ana.	Viz.	Report	Pres.
Ibrahim	✓	✓	✓	✓	✓	✓	✓	✓
Haseeb	✓	✓					✓	✓
Asadohoo							✓	✓
Sameer							✓	✓
Ayaan							✓	✓
Pak Hei Ng					✓	✓	✓	

Legend: Data = data collection; Proc. = processing and feature engineering; NLP = embedding, clustering, and labelling; Model = training and inference; Ana. = statistical analysis; Viz. = plots and visual diagnostics; Report = writing and editing; Pres. = presentation preparation.

Member contributions (brief).

1. **Ibrahim:** Led end-to-end implementation across the pipeline, including data collection/processing, NLP enrichment, modeling, analysis, visualizations, and final write-up and presentation materials.
2. **Haseeb:** Contributed to data processing and feature/pipeline work, supported core implementation tasks, and helped refine the report and presentation.
3. **Asadohoo:** Focused on report writing/editing and supporting the final presentation deliverables.
4. **Sameer:** Contributed primarily to report writing/editing and presentation preparation.
5. **Ayaan:** Contributed primarily to report writing/editing and presentation preparation.
6. **Pak Hei Ng:** Drove analysis and visualization work, supported report writing, and helped interpret results and figures.

References

[1] United Nations Department of Economic and Social Affairs. International migration 2020 highlights. Report ST/ESA/SER.A/404, United Nations, 2020.

[2] U.S. Customs and Border Protection. Southwest land border encounters. https://www.cbp.gov/newsroom/stats/southwest-land-border-encounters, 2026. Fiscal Years 2019–2026 (year-to-date).

[3] U.S. Department of State, Bureau of Consular Affairs. Monthly immigrant visa issuance statistics. https://travel.state.gov/content/travel/en/legal/visa-law0/visa-statistics/immigrant-visa-statistics/monthly-immigrant-visa-issuances.html, 2025. Accessed: January 2017 – September 2025.

[4] Ritchie Vink. Polars: Blazingly fast DataFrames in Rust and Python. https://pola.rs, 2024.

[5] Google. Google news RSS feed. https://news.google.com, 2025. Accessed via pygooglenews RSS API.

[6] Adrien Barbaresi. Trafilatura: A web scraping library and command-line tool for text discovery and extraction. https://trafilatura.readthedocs.io, 2021.

[7] Google. Google trends. https://trends.google.com, 2025. Accessed via pytrends API.

[8] International Monetary Fund. Real effective exchange rate (REER). https://data.imf.org, 2025. Accessed via IMF Data portal.

[9] Michael Günther, Louis Milliken, Jonathan Geuter, Georgios Mastrapas, Bo Wang, and Han Xiao. Jina embeddings: A novel set of high-performance sentence embedding models. *arXiv preprint arXiv:2307.11224*, 2024.

[10] NVIDIA Corporation. TensorRT: High-performance deep learning inference optimizer and runtime. https://developer.nvidia.com/tensorrt, 2024.

[11] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD)*, pages 160–172. Springer, 2013.

[12] RAPIDS Development Team. RAPIDS cuML: GPU-accelerated machine learning. https://rapids.ai, 2024.

[13] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

[14] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

[15] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[16] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.

[17] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[18] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[19] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[20] Peter J. Huber. Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–101, 1964.

[21] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. https://pytorch.org, 2019.

[22] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 5998–6008, 2017.

[23] James D. Hamilton. *Time Series Analysis*. Princeton University Press, 1994.

[24] Christopher A. Sims. Macroeconomics and reality. *Econometrica*, 48(1):1–48, 1980.

Appendices

A Data and Pipeline Engineering (Reproducible Overview)

This appendix provides the technical reference for reproducing the end-to-end system and interpreting its outputs. The pipeline transforms raw monthly data streams into multi-horizon forecasts via six stages:

Collection → Processing → NLP Enrichment → Panel Construction → Training → Inference

Data sources (monthly cadence). The system integrates four streams: (i) U.S. Department of State visa issuances, (ii) CBP Southwest land border encounters, (iii) Google News articles collected by country and topic, and (iv) Google Trends keyword time series, with IMF REER exchange rate data available for a subset of countries.

News collection engineering. The news pipeline was engineered for throughput and fault tolerance:

- Articles were collected via the Google News RSS API [5], with encoded URLs resolved through a batched decoding endpoint (20 articles per batch), reducing API calls by ~95%.
- Full article text was fetched asynchronously with semaphore-controlled concurrency (max 16 connections), exponential backoff with jitter, and a 30-second watchdog timeout per request.
- HTML-to-text extraction used Trafilatura [6], parallelized across CPU cores with a `ProcessPool Executor`.
- Intermediate results were checkpointed every 25 articles via atomic writes (write to temp file, then rename), enabling crash-resilient resumption.

Reproducibility (recommended workflow). We provide two ways to reproduce the results. The recommended path is to use the project's documented bootstrap routine to download the processed datasets and model artifacts from Hugging Face into the expected directory structure. For completeness, the repository also supports a live collection workflow that re-downloads raw inputs from source systems and then re-runs processing, training, and analysis.

```
# Install dependencies (recommended: uv) and bootstrap artifacts
uv sync
source .venv/bin/activate
python -m src.main bootstrap --org sdsc2005-migration

# Validate the bootstrap plan without downloading (dry run)
python -m src.main bootstrap --org sdsc2005-migration --dry-run

# Rebuild the training panel and retrain models
python -m src.processing.build_panel
python -m src.models.train_and_evaluate
```

If `uv` is not available, an equivalent local setup can be created with a virtual environment and `pip` (see `requirements.txt` and the `README.md`). For additional control over downloads, the repository also provides a Hugging Face sync utility that can download default datasets and model artifacts into local folders.

Plots produced during exploratory analysis and evaluation are stored under `reports/plots/`, organized into `eda_plots/`, `cluster_plots/`, `model_performance/`, and `trends_vs_migration/`.

B Panel Schema and Feature Dictionary

All model training and inference operate on a monthly country-level panel constructed from lagged signals with forward-looking lead targets.

Lag features (18). For each of three signals, six months of history are included:

- Visa volume: `visa_lag_1` through `visa_lag_6`
- Exchange rate: `exchange_rate_lag_1` through `exchange_rate_lag_6`
- News events: `news_events_lag_1` through `news_events_lag_6`

Lead targets (6). The prediction targets are `target_visa_lead_{1..6}`.

Missing data policy. Exchange-rate values are available for 6 of 15 countries; unavailable values are zero-filled to avoid injecting spurious trends. Months with no collected news for a country are assigned zero event counts. Observations without sufficient lag history (first six months per country) are dropped.

C Per-Horizon Surge Classification

The summary Table 2 in the main body reports averages across horizons. Tables 5–8 provide the per-lead breakdown for each model on the out-of-time test set (approximately 103–114 surge instances; threshold $> 1.5\sigma$). The decline from Lead 1 to Lead 6 reflects the increasing difficulty of long-range forecasting; the ensemble reduces this decay relative to any single architecture.

Table 5: Random Forest surge classification (OOT).

Horizon	Prec.	Rec.	F1
Lead 1	0.96	0.97	**0.97**
Lead 2	0.93	0.95	**0.94**
Lead 3	0.88	0.96	**0.92**
Lead 4	0.85	0.94	**0.89**
Lead 5	0.82	0.93	**0.88**
Lead 6	0.78	0.91	**0.84**

Table 6: Transformer Encoder surge classification (OOT).

Horizon	Prec.	Rec.	F1
Lead 1	0.95	0.93	**0.94**
Lead 2	0.95	0.94	**0.94**
Lead 3	0.91	0.93	**0.92**
Lead 4	0.86	0.90	**0.88**
Lead 5	0.85	0.90	**0.88**
Lead 6	0.83	0.90	**0.87**

Table 7: LSTM (SurgeJointLoss) surge classification (OOT).

Horizon	Prec.	Rec.	F1
Lead 1	0.97	0.85	**0.91**
Lead 2	0.94	0.84	**0.89**
Lead 3	0.91	0.83	**0.87**
Lead 4	0.88	0.81	**0.84**
Lead 5	0.85	0.80	**0.82**
Lead 6	0.82	0.79	**0.80**

Table 8: Horizon-Aware Ensemble surge classification (OOT).

Horizon	Prec.	Rec.	F1
Lead 1	0.96	0.96	**0.96**
Lead 2	0.93	0.96	**0.95**
Lead 3	0.92	0.94	**0.93**
Lead 4	0.88	0.94	**0.91**
Lead 5	0.83	0.94	**0.88**
Lead 6	0.80	0.92	**0.86**

D Ensemble Weight Schedule

Table 9 lists the horizon-specific weighting scheme used to combine base model predictions. Weight mass shifts from Random Forest at short horizons to the Transformer at longer horizons, reflecting each architecture's comparative advantage by lead time.

Table 9: Horizon-Aware Ensemble weights (sum to 1.0 per row).

Lead	RF	Trans.	LSTM	Rationale
1 mo	60%	10%	30%	RF and LSTM dominate short-term
2 mo	50%	20%	30%	RF remains dominant
3 mo	30%	40%	30%	Balanced transition
4 mo	20%	60%	20%	Transformer becomes critical
5 mo	10%	80%	10%	Transformer dominates long horizon
6 mo	5%	85%	10%	Transformer captures slow trends

E Detailed Event Signal Analysis

Of 392 country–event–lag combinations screened, 58 (14.8%) survived Benjamini–Hochberg correction at $q \leq 0.10$. Selected highlights across modalities are described below.

News event clusters. Among cluster-level signals, Mexico's geopolitical-watch cluster showed $r = -0.436$ at a 1-month lag ($q = 0.0009$), India's economic-sentiment cluster showed $r = -0.424$ at 2 months ($q = 0.0010$, surge lift 0.932), Honduras's Evangelical-pastoral cluster showed $r = +0.4250$ at 2 months ($q = 0.0114$, lift 1.229), Vietnam's misinformation cluster showed $r = -0.403$ at 5 months ($q = 0.0017$, lift 0.816), El Salvador's conflict cluster showed $r = -0.361$ at 5 months ($q = 0.0055$, lift 0.842), Guatemala's "Border Policy" cluster showed $r = +0.523$ at 3 months, and Venezuela's "Economic Crisis" cluster showed $r = +0.509$ at 5 months. Eleven of the 15 countries exhibited at least one significant event signal.

Headline sentiment scores. Aggregate headline-level scores revealed sharper relationships still. The Dominican Republic produced the single strongest lead in the study: $r = 0.617$ at a 3-month lag ($q = 3.3 \times 10^{-9}$). Cuba's "Trump Policy" cluster showed $r = -0.595$ at lag 0 ($q = 3.4 \times 10^{-9}$), capturing the immediate visa impact of U.S. policy shifts.

Exchange rate and Trends signals. Among the 6 countries with REER data, the Dominican Republic again dominated ($r = 0.498$, lag 2, $q = 2.6 \times 10^{-6}$), followed by Mexico ($r = 0.289$, lag 6, $q = 0.018$). Google Trends was significant for 66 of 120 rows (55.0%); notable examples include China's `us_visa` keyword ($r = 0.406$, $q = 1.1 \times 10^{-4}$) and Afghanistan's `us_jobs` ($r = -0.484$, $q = 4.4 \times 10^{-6}$).

Table 10 ranks the ten strongest significant signals across all modalities and countries.

Table 10: Top-10 significant lead–lag signals ranked by $|r|$ (BH-corrected, $q \leq 0.10$).

Country	Signal / Cluster	Lag	Pearson r	q-value	Modality
Dominican Republic	"First Presidency" sentiment	3 mo	+0.617	3.3×10^{-9}	Sentiment
Cuba	"Trump Policy" events	0 mo	−0.595	3.4×10^{-9}	Event cluster
Guatemala	"Border Policy"	3 mo	+0.523	significant	Event cluster
Venezuela	"Economic Crisis"	5 mo	+0.509	significant	Event cluster
Dominican Republic	REER exchange rate	2 mo	+0.498	2.6×10^{-6}	Exchange rate
Afghanistan	`us_jobs` Trends	6 mo	−0.484	4.4×10^{-6}	Google Trends
Mexico	"Geopolitical Watch"	1 mo	−0.436	9.4×10^{-4}	Event cluster
Honduras	"Evangelical Pastors"	2 mo	+0.425	1.1×10^{-2}	Event cluster
India	"Economic Sentiment"	2 mo	−0.424	1.0×10^{-3}	Event cluster
China	`us_visa` Trends	3 mo	+0.406	1.1×10^{-4}	Google Trends

F Cluster Quality Analysis

Figures 7 and 8 summarise the quality and thematic granularity of the HDBSCAN event clusters produced for each of the 15 origin countries.

Silhouette Scores. A silhouette score above 0.5 indicates well-separated, internally coherent clusters. As Figure 7 shows, the majority of countries achieve scores above this threshold; the United States (US-centric context articles), China, and the Philippines reach 0.70–0.82, confirming that Jina v5 embeddings capture thematically distinct news groupings. Cuba and Colombia are the notable exceptions (scores 0.28–0.43), reflecting noisier or more topically overlapping news coverage for those origins.

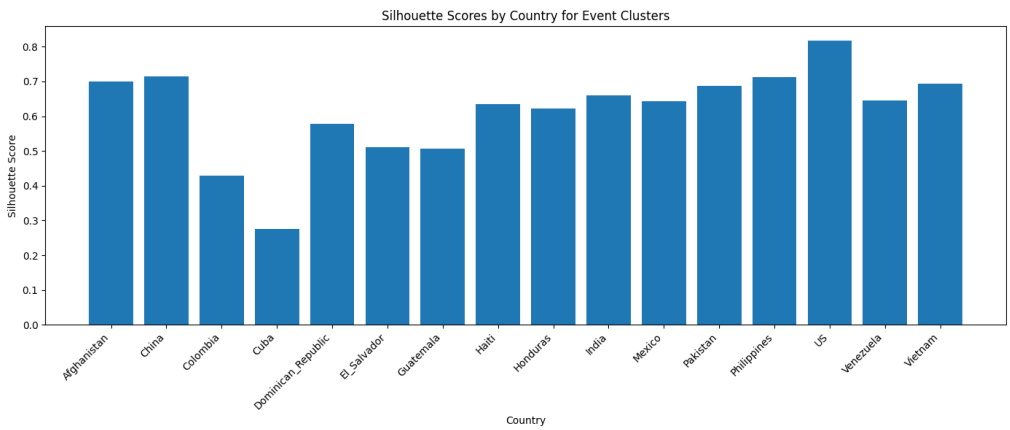

Figure 7: Silhouette scores by country for HDBSCAN event clusters. Values above 0.5 indicate well-separated clusters. Most countries exceed this threshold, supporting the embedding space as a basis for thematic grouping.

Number of Unique Clusters. Figure 8 shows that Cuba, El Salvador, and Guatemala produced the largest number of distinct event themes (6 clusters each), reflecting richer and more varied news landscapes for these countries in the context of U.S. migration. Countries with fewer clusters (3) such as Afghanistan, Colombia, and India tended to have more concentrated coverage around a single dominant narrative (e.g., Taliban governance or H-1B visa policy), which can make cluster labels simpler but potentially reduces signal diversity.

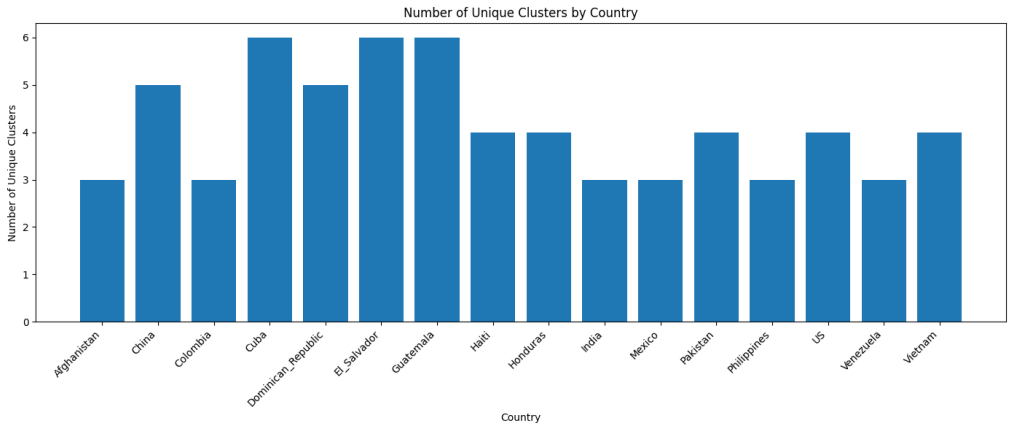

Figure 8: Number of unique HDBSCAN event clusters per country. Greater cluster counts indicate richer thematic diversity in the collected news corpus, providing more candidate leading indicators per origin.

G F1-Score Trajectory Across Forecast Horizons

Figure 9 plots per-lead F1 scores for all four models on the OOT test set, complementing the tabular summaries in Tables 5–8 and giving an at-a-glance view of how each architecture's surge-detection ability degrades with forecast distance.

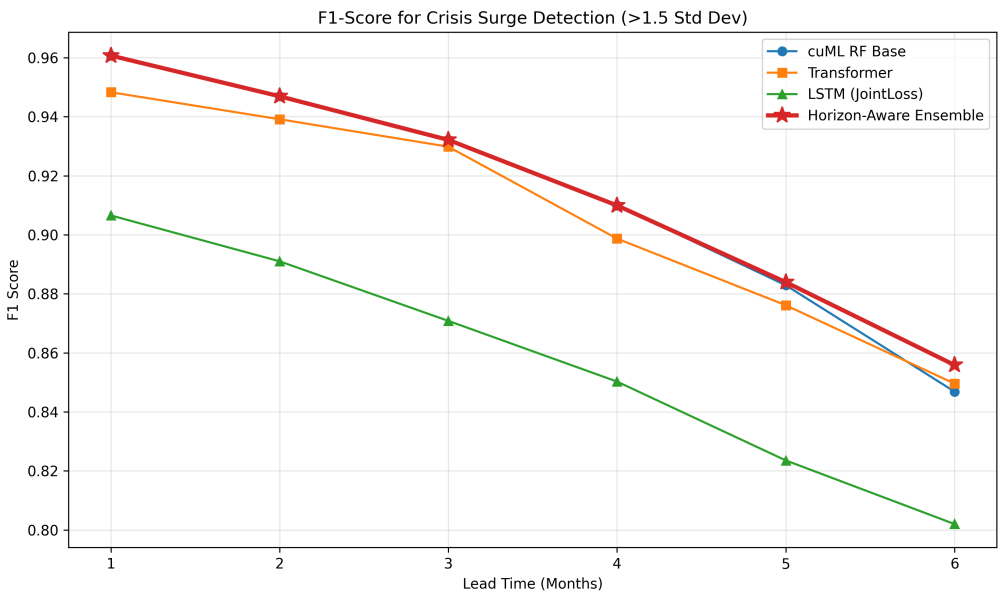

Figure 9: F1 scores for surge detection across forecast lead times (Leads 1–6). All models degrade with increasing horizon; the Horizon-Aware Ensemble (red star) consistently outperforms individual architectures at every lead. The LSTM (JointLoss, green triangle) trades lower F1 for higher precision, visible as the most conservative trajectory.

The ensemble curve is notably flatter than that of any individual model, confirming that the horizon-specific weighting schedule (Appendix D) successfully compensates for the complementary decay profiles of Random Forest and Transformer. The Horizon-Aware Ensemble maintains F1 > 0.86 even at Lead 6, a 2-percentage-point improvement over the best single model at that horizon.

H Future Work

1. **Social media integration:** Incorporate social platforms and messaging signals to capture migration intent at finer temporal resolution than monthly news coverage.
2. **Cross-country transfer learning:** Transfer representations from data-rich countries (e.g., Mexico, India) to data-sparse origins (e.g., Haiti, Afghanistan) to reduce cold-start error.
3. **Higher-frequency inference:** Extend the batch-monthly pipeline to rolling weekly forecasts where upstream data availability permits.
4. **Richer macro indicators:** Add inflation, unemployment, remittances, and commodity prices to reduce reliance on partial REER coverage.
5. **Causal validation:** Use Granger-style tests, structural models, and natural-experiment designs to probe whether leading indicators are causal mechanisms or correlated symptoms.